

Origin in Tobacco Smoke of *N*'-Nitrosonornicotine, a Tobacco-Specific Carcinogen: Brief Communication^{1,2,3}

Dietrich Hoffmann,⁴ Michael Dong,⁴ and Stephen S. Hecht^{4,5,6}

ABSTRACT—To evaluate risk factors and to approach methods of reduction of the carcinogenic potential of cigarette smoke, the transfer rate of *N*-nitrosonornicotine in a popular U.S. blended cigarette into mainstream smoke was quantitatively determined. The mean transfer rate was 11.3%; thus ~46% of the tobacco-specific carcinogen in the smoke came from the tobacco, and the remainder was synthesized during smoking.—*J Natl Cancer Inst* 58: 1841-1844, 1977.

Tobacco chewing and smoking are associated with cancer of the oral cavity and esophagus (1-5). NNN has been identified in tobacco products and shown to be an esophageal carcinogen in rats (6). Concentrations range from 0.3 to 90 ppm in tobacco (7-10) and from 140 to 250 ng in the smoke of one cigarette (11). These levels are high compared to nitrosamine concentrations in meat and other food products, which rarely exceed 0.1 ppm (12). NNN was not detected in freshly harvested tobacco (<1 ppb) but was formed during tobacco processing (13). The presence of NNN in smoke can be explained either by its direct transfer from tobacco into smoke and/or by its formation during smoking. If a significant portion of NNN in cigarette smoke originated by transfer from tobacco, appropriate steps could then be taken to inhibit its formation during tobacco curing and processing.

MATERIALS AND METHODS

¹⁴C-Labeled NNN (14) was used as a radioactive tracer for NNN transfer into mainstream smoke, and [³H]NNN was used as the internal standard. The analytic procedure is a modified version from an earlier published method (15).

Apparatus.—A Packard 3385 Tri-Carb was used for the scintillation counting. Discriminator settings were optimized for the most efficient simultaneous counting of ¹⁴C and ³H labels in a scintillation cocktail of 0.4% of PPO (2,5-diphenyloxazole) and 0.005% POPOP (1,4-bis[2-(5-phenyloxazolyl)]benzene) in toluene. For analysis, we used a model ALC/GPC-202 high-speed liquid chromatograph (Waters Associates, Inc., Milford, Mass.) with a 254-nm differential ultraviolet detector, a Hewlett-Packard gas liquid chromatograph 5710 with a flame ionization detector, and a Hewlett-Packard model 5980A mass spectrometer. The cigarettes were smoked with a Borgwaldt 20-port smoking machine, on which every second smoking channel of the rotating head was connected to a nitrogen source (flow rate, 17.5 ml/sec) (7, 11).

Reagents.—All solvents were spectrograde. [2-¹⁴C]NNN (4.1 mCi/mole) and unlabeled NNN were synthesized as described previously (14). Basic alumina (activity II-III ICN-Pharmaceuticals, Inc., Cleveland, Ohio) was used for chromatography. Carbowax 20-M TPA (10%) on Chromosorb W (mesh, 80-100) was ob-

tained from Applied Science Laboratories, Inc. (State College, Pa.). [³H]Nornicotine (45 mg, 0.30 mmole, 296 mCi), obtained by catalytic tritium exchange (New England Nuclear Corp., Boston, Mass.) of nornicotine, was dissolved in 3.0 ml of 1 N HCl. After dropwise addition of 1 ml solution of NaNO₂ (700 mg, 10.1 mmole), the resulting mixture was allowed to stand at room temperature for 24 hours; it was then brought to pH 12 with aqueous NaOH and extracted three times with 10 ml of CHCl₃, dried (Na₂SO₄), and concentrated. It yielded a residue which was purified by preparative thin-layer chromatography on silica gel with elution by chloroform-methanol (9:1) (*R_f* of NNN = 0.60). This [³H]-NNN (42% yield; >99% purity according to GLC analysis) had a specific activity of 260 mCi/mole. In order to obtain material of constant specific activity with respect to the analytic procedure, a mixture of [³H]NNN and nicotine was carried through the entire analytic process as described below, and the [³H]NNN was isolated in the final step by HPLC. This procedure was repeated to give [³H]NNN a constant specific activity (210 mCi/mole), which was then used as internal standard in the transfer rate study of [¹⁴C]NNN.

Cigarettes.—The nonfilter cigarettes (85-mm) were bought on the open market in Westchester County, New York in the winter of 1975-76. Two hundred cigarettes were opened, and 20-g samples of their tobacco (moisture content, 11.0%) were used for tobacco analysis. For the smoke analysis of cigarettes treated with [¹⁴C]NNN 50 weight-selected cigarettes (±20 mg of average weight of 200 cigarettes) were used in one experiment; in all other experiments, 100 non-weight-selected cigarettes were smoked. The standard smoking conditions were 1 puff/minute of 35-ml volume and 2 seconds' duration, to a butt length of 23 mm (16, 17).

Cigarettes labeled with [¹⁴C]NNN.—A solution of [¹⁴C]NNN in water (0.017 mg/ml) was prepared for injection with an automatic microsyringe applicator using a 50-μl Hamilton syringe. The microswitch settings of the instrument were such that 38 μl (0.66 μg [¹⁴C]NNN; 3.4 × 10⁴ dpm) of the solution was deposited in the mid-

ABBREVIATIONS USED: NNN = *N*'-nitrosonornicotine; GLC = gas liquid chromatography; HPLC = high-pressure liquid chromatography.

¹ Received August 4, 1976; accepted November 29, 1976.

² Supported by Public Health Service contract N01-CP55666 from the Division of Cancer Cause and Prevention, National Cancer Institute (NCI).

³ No. LII of "Chemical Studies on Tobacco Smoke."

⁴ Division of Environmental Carcinogenesis, Naylor Dana Institute for Disease Prevention, American Health Foundation, Valhalla, N.Y. 10595.

⁵ Recipient of Public Health Service Career Development Award 1-K04-CA00124 from the NCI.

⁶ We thank W. Z. Maddox and M. R. Guerin, Analytical Chemistry Division, Oak Ridge National Laboratory, Oak Ridge, Tennessee, for supplying an "Automatic Syringe Applicator for Cigarettes."